

INTRODUCTION

- Tracks **defects** cause rail accidents.
- Manual checks are **time consuming**.
- AI robot with camera + **GPS defects faults**.
- Goal : **safer, faster rail operations**.
- **Affordable and easy to deploy**.
- **Reduces human efforts and error**

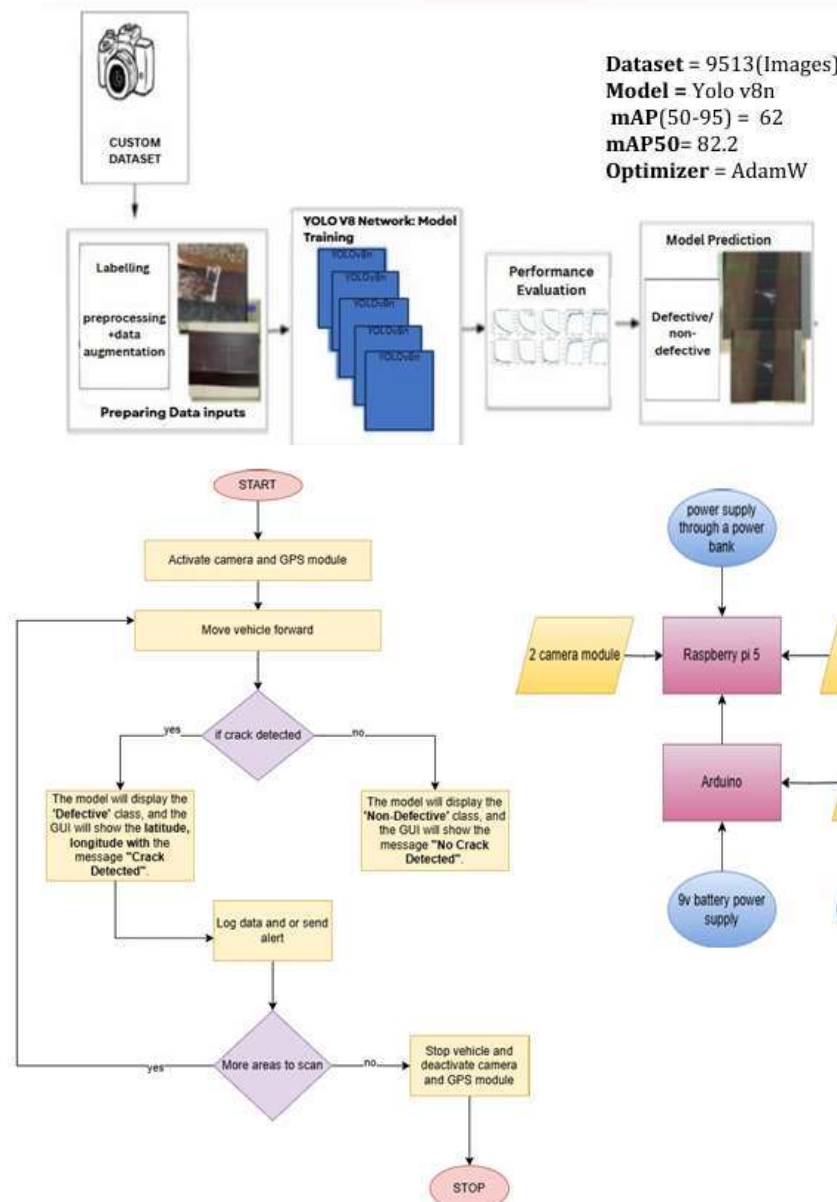


OBJECTIVES

1. To implement a **crack detecting algorithm** that can accurately **detect and track on railway tracks** using video frames.
2. To develop a **cost-effective prototype** for railway track crack detection.
3. **Integrating GPS module** for location updates with GUI.



METHODOLOGY



RESULTS



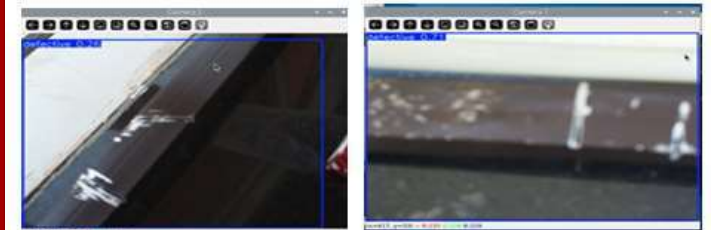
Railway crack detection bot and environment prototype

?? Live Crack Detection Feed

Time	Status	Latitude	Longitude
2023-09-13T12:44:16.7	Crack Detected	18.5435	73.8122
2023-09-13T12:44:18.1	Crack Detected	18.5435	73.8122
2023-09-13T12:44:19.5	Crack Detected	18.5435	73.8122
2023-09-13T12:44:20.9	Crack Detected	18.5435	73.8122
2023-09-13T12:44:22.3	Crack Detected	18.5435	73.8122
2023-09-13T12:44:23.7	Crack Detected	18.5435	73.8122
2023-09-13T12:44:25.1	Crack Detected	18.5435	73.8122
2023-09-13T12:44:26.5	Crack Detected	18.5435	73.8122
2023-09-13T12:44:27.9	Crack Detected	18.5435	73.8122
2023-09-13T12:44:29.3	Crack Detected	18.5435	73.8122
2023-09-13T12:44:30.7	Crack Detected	18.5435	73.8122
2023-09-13T12:44:32.1	Crack Detected	18.5435	73.8122
2023-09-13T12:44:33.5	Crack Detected	18.5435	73.8122
2023-09-13T12:44:34.9	Crack Detected	18.5435	73.8122
2023-09-13T12:44:36.3	Crack Detected	18.5435	73.8122

Export CSV Crack Detected

This is a GUI displaying location of the crack detected



Camera feed showing defective region

FUTURE SCOPE

- Add **sensor fusion (LiDAR, Ultrasonic, IR)** for better accuracy.
- Send **real-time alerts** to authorities.
- **Detect more defects** (e.g., broken rails, missing parts).
- Use **Solar power** for outdoor use.
- Expand coverage to tunnels and bridges.

Name, students and Guide/s

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